# Markscheme

Mathematics: analysis and approaches
Practice question bank

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**132. (a)**

The complex number $z$ satisfies $|z| = 5$ and $\mathrm{Re}(z) = 3$, with $\mathrm{Im}(z) < 0$.

Find $\bar{z} \times i$ in the form $a + bi$.

$\mathrm{Im}(z)^2 = 25 - 9 = 16 \Rightarrow \mathrm{Im}(z) = -4$ (since negative) *(M1)*

$$z = 3 - 4i \Rightarrow \bar{z} = 3 + 4i$$ **A1**

$$i(3 + 4i) = 3i + 4i^2$$ *(M1)*

$$= -4 + 3i$$ **A1**

**[5 marks]**